

SMD LED



Lead-Free Parts

LG-1206HEL-CT-A

DATA SHEET

DOC.NO : IMQW0905-LG-1206HEL-CT-A

REV. : A

DATE : 17 – Oct. – 2022

Features:

1. Package in 8mm carrier tape on 7" diameter reel.
2. Compatible with automatic placement equipment.
3. Compatible with infrared and vapor phase reflow solder process.
4. Mono-color type.
5. Pb-free.
6. The product itself will remain within RoHS compliant version.

Descriptions:

1. The LG-1206 SMD Taping is much smaller than lead frame type components, thus enable smaller board size, higher packing density, reduced storage space and finally smaller equipment to be obtained.
2. Besides, lightweight makes them ideal for miniature applications. etc.

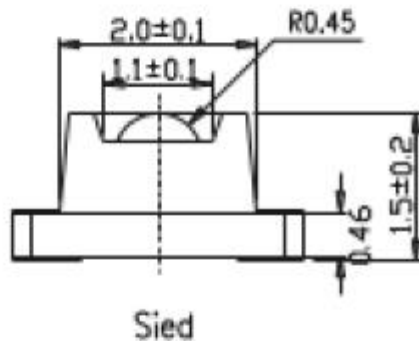
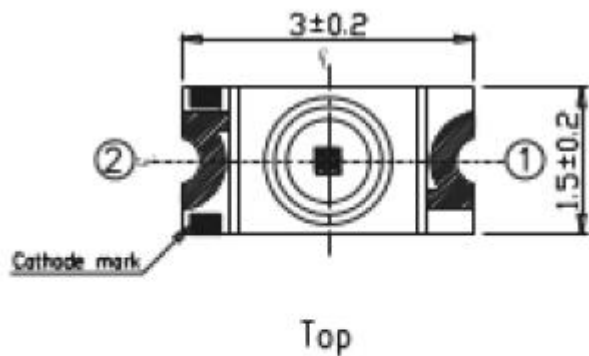
Applications:

1. Backlighting in dashboard and switch.
2. Telecommunication : indicator and backlighting in telephone and fax.
3. Flat backlight for LCD, switch and symbol
4. General use.

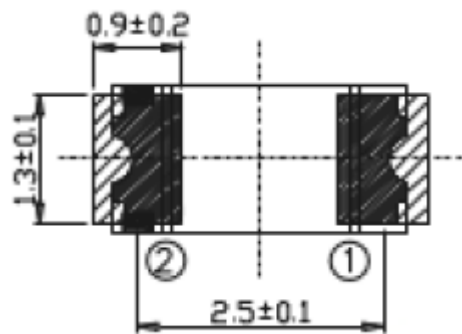
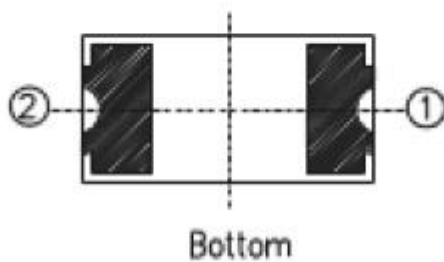
Device Selection Guide

Part No.	Material	COLOR	
		Emitted	Lens
LG-1206HEL-CT-A	AlGaInP	Red	Water Clear

Package Outline Dimensions



Recommended Soldering Pad Dimensions



Unit:mm

- Note: 1. All dimension are in millimeter tolerance is ± 0.1 mm unless otherwise noted.
2. Specifications are subject to change without notice.

Absolute Maximum Ratings at Ta=25°C

Parameter	Symbol	Ratings	UNIT
Power Dissipation	PD	60	mW
Peak Forward current Duty 1/10@1KHz	I _{FP}	50	mA
Forward Current Per Chip	I _F	25	mA
Electrostatic Discharge	ESD	2000	V
Operating Temperature	T _{opr}	-40 ~ +85	°C
Storage Temperature	T _{stg}	-40 ~ +90	°C
Soldering Temperature	T _{sol}	Reflow Soldering : 260 °C for 10 sec. Hand Soldering : 350 °C for 3 sec.	

Typical Electrical & Optical Characteristics (Ta=25°C)

Parameter	Symbol	Min.	Typ.	Max.	Unit	Condition
Luminous Intensity	IV	140	----	285	mcd	IF=20mA
Peak Wavelength	λ _p	----	632	----	nm	
Dominant Wavelength	λ _d	617.5	----	633.5	nm	
Spectrum Radiation	△λ	----	20	----	nm	
Forward Voltage	VF	1.75	----	2.35	V	
Viewing Angle	2θ1/2	----	60	----	Deg	
Reverse Current	I _r	----	----	10	μA	V _r =5V

Note : 1.The forward voltage data did not including ±0.1V testing tolerance.
2.The luminous intensity data did not including ±11% testing tolerance.

Luminous Intensity Classification

IV (mcd) at 20mA		
BIN CODE	Min	Max
R2	140	180
S1	180	225
S2	225	285

Dominant Wavelength Classification

λd (nm) at 20mA		
BIN CODE	Min	Max
E4	617.5	621.5
E5	621.5	625.5
E6	625.5	629.5
E7	629.5	633.5

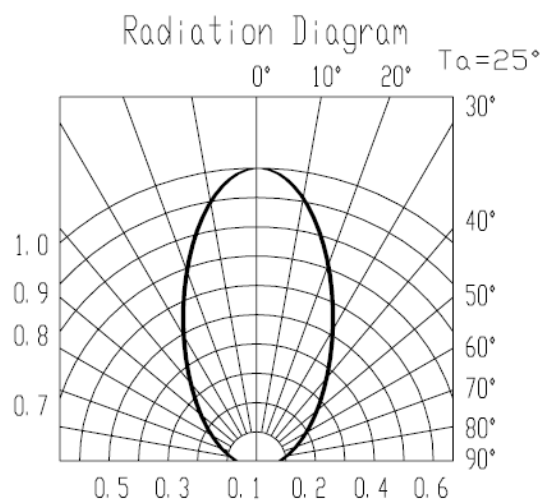
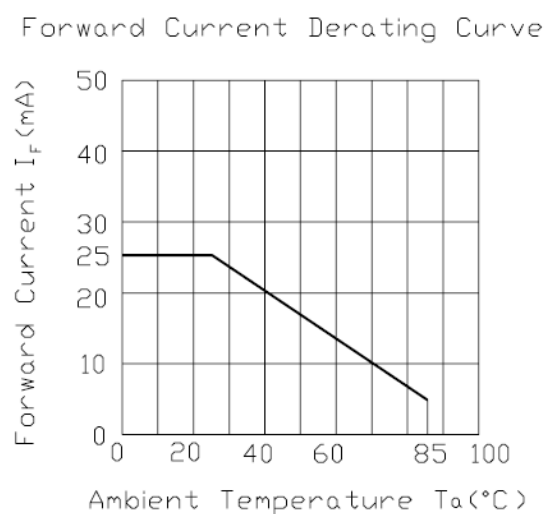
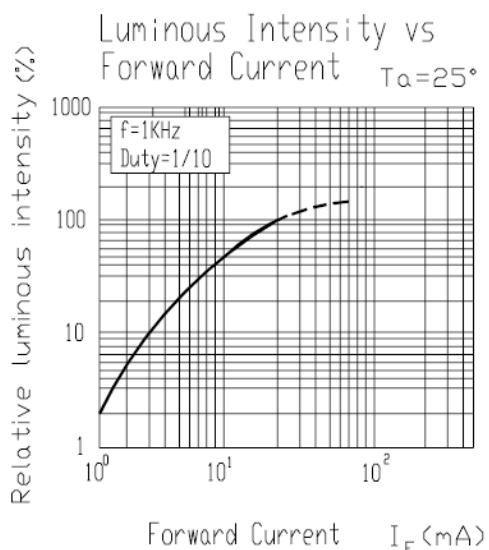
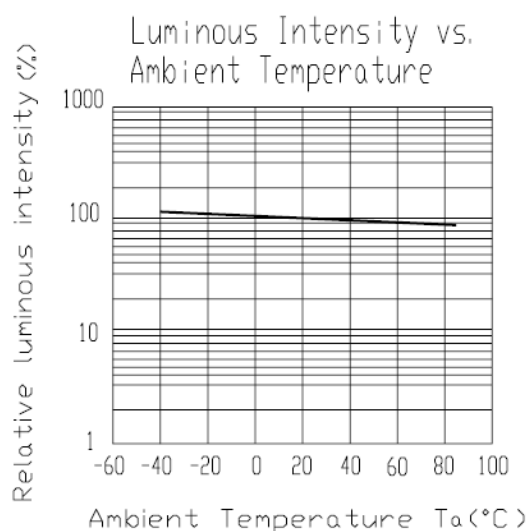
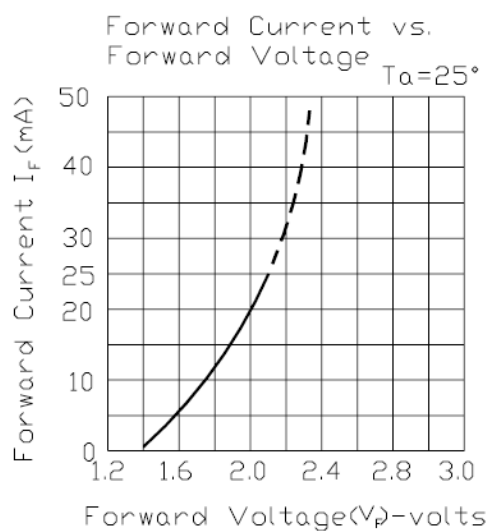
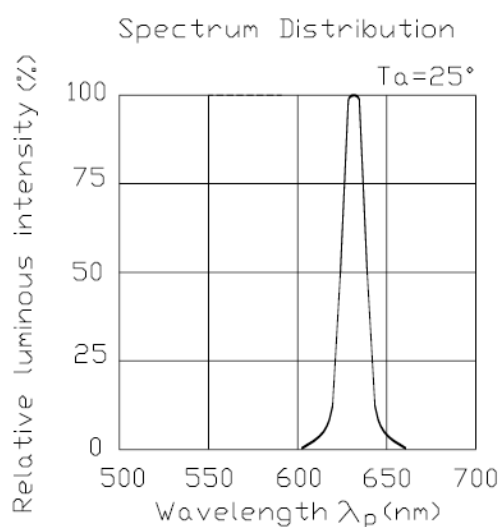
Forward Voltage Classification

VF(V) at 20mA		
BIN CODE	Min	Max
0	1.75	1.95
1	1.95	2.15
2	2.15	2.35

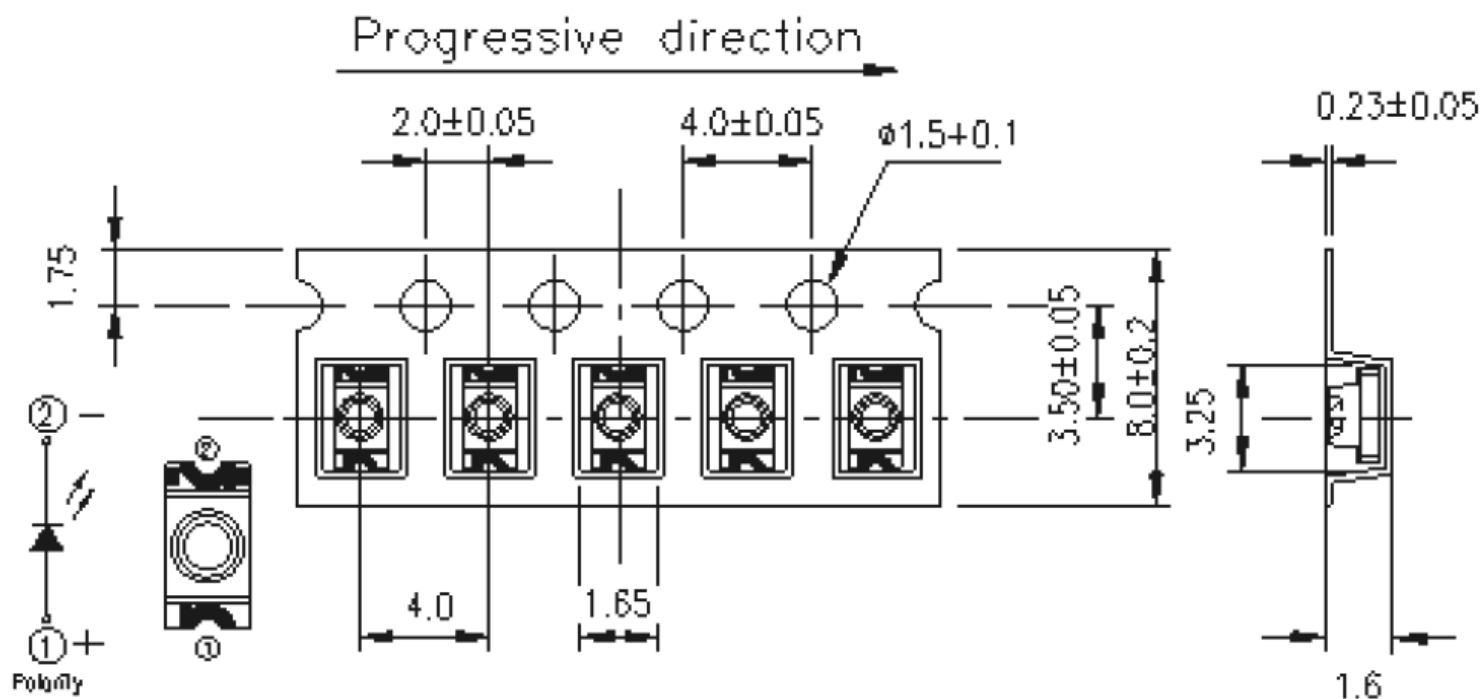
Note:

- 1:Tolerance of Luminous Intensity $\pm 11\%$
- 2:Tolerance of Forward Voltage $\pm 0.1V$
3. Tolerance of The Dominant Wavelength $\pm 1nm$

Typical Electro-Optical Characteristics Curve

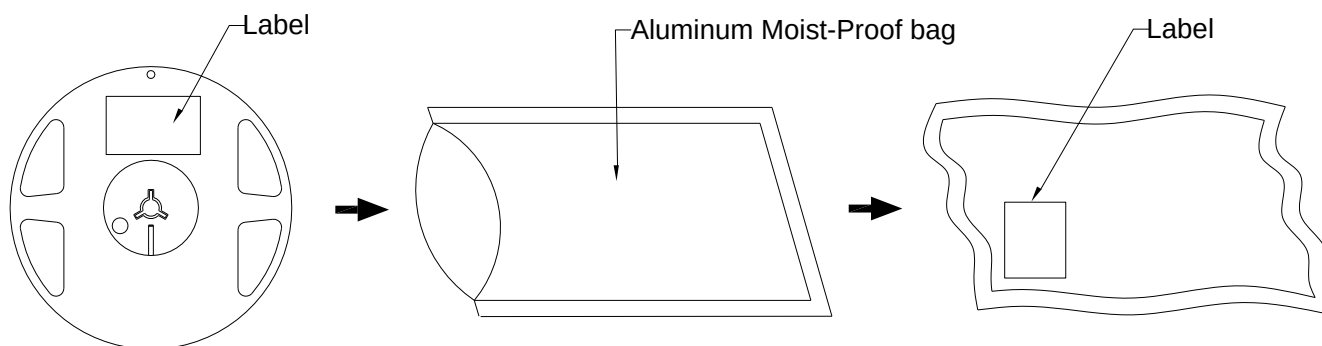


Carrier Tape Dimensions



Note : The tolerances unless mentioned is ± 0.1 mm, Angle ± 0.5 . Unit=mm.

Packing Specifications



Part No.	Description	Quantity/Reel
LG-1206HEL-CT-A	8.0mm tape, 7" reel	3000 devices

Label Explanation

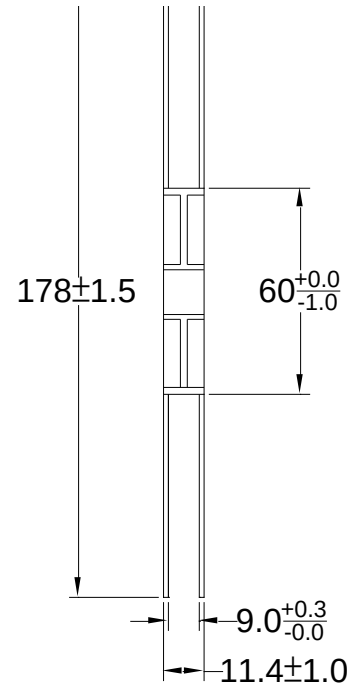
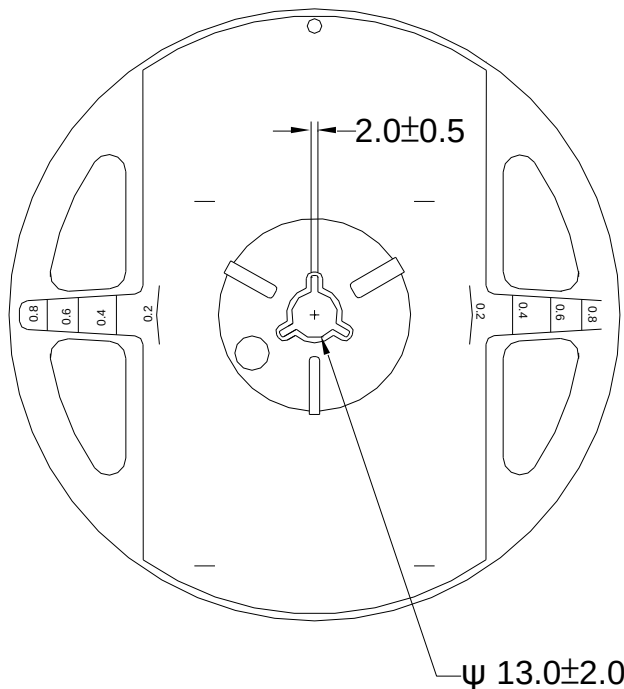
	LIGITEK ELECTRONICS CO., LTD.
PART :	XXXXXXXXXXXXXXXXXXXX
LOT :	XXXXXXXXXX
QTY(PCS):	XXXX
BIN/HUE :	XXXXX
	VF: XX-XX

BIN : Luminous Intensity

HUE : Dominant Wavelength

VF: Forward Voltage

Reel Dimensions



Precautions For Use:

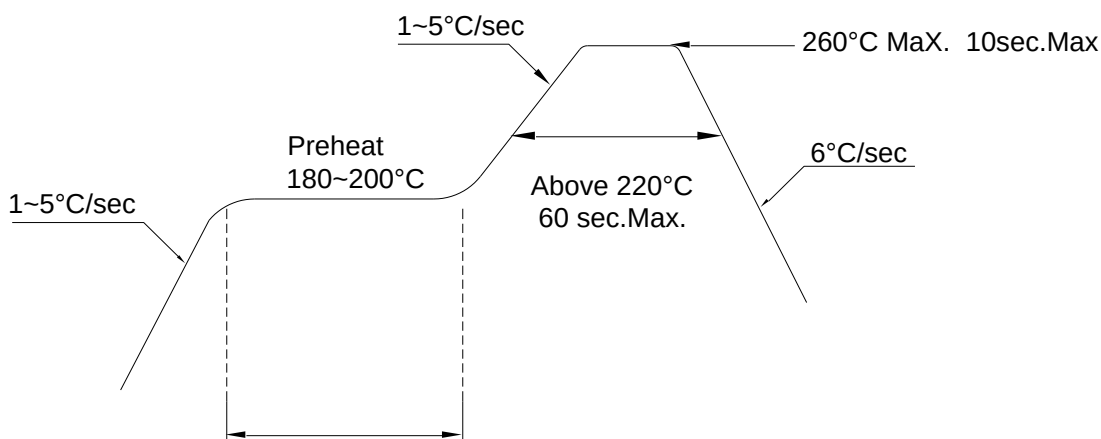
Storage time:

1. Calculated shelf life before opening is 18 months at $< 30^{\circ}\text{C}$ and $< 90\%$ relative humidity (RH)
2. After bag is opened, devices which will be subjected to reflow soldering or other high temperature processes must be
 - a) Assembled within one years in an environment of $\leq 30^{\circ}\text{C}$ / 60% RH, or
 - b) Stored at ambient of 10% RH or less
3. Devices are required baking before assembly if:
 - a) Humidity Indicator Card reads $>10\%$ (for level 2a -5a) or $>60\%$ (for level 2) at ambient temperature $23\pm 5^{\circ}\text{C}$
 - b) 2.a) or 2.b) doesn't meet
4. If baking is required, devices should be baked for >72 hours at $60\pm 5^{\circ}\text{C}$ / 5% RH. Performing baking only once, and using the baked devices within 72 hours.

Over current- protection

The LEDs is sensitive parts, slight voltage shift will cause big change and will cause burn out. Customer must apply resistors for protection.

LED soldering



Note:

1. Reflow soldering should not be done more than two times.
2. When soldering, do not put stress on the LEDs during heating.
3. After soldering, do not warp the circuit board.

Repairing

In principle repair should not be done after the LEDs have been soldered. When repairing is unavoidable, it should be confirmed before hand not to be damaged whether the characteristics of the LEDs by repairing and a double-head soldering iron should be used (as below description figure).

